PAPER_1293_SMALE_13TH_HILBERT_16TH_SIMPLIFIED

Daniel T. Murphy

PAPER_1293 — Smale 13th — Hilbert 16th Simplified

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: A — Mathematical Conjecture (CLOSED)
Date: June 11, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — UQFF integer-primitive derivation
Calculator surface: `calculate_paradox({"paradox": "smale_13th"})`
Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1293})`

Master Expression

```
Simplified to low-degree polynomials, same K_MEX × n²/2 bound applies
```

Match: ROUTING

Physical / Mathematical Interpretation

Smale's 13th problem is Hilbert's 16th restricted to low-degree polynomial vector fields. Same bound.

UQFF Locked Primitives Used

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$
- $\rho_{SCm} = 7.09 \times 10$■^{3}■ J/m^3, $\omega_{SCm} = 1.25$ THz, $S_{26} = 1.453162$

Live Calculator Verification

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "smale_13th"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_smale_13th_closure`
- Dispatch: `PARADOX_TO_CLOSURE["smale_13th"]`
